

1 Example 1

1.1 Problem description

A technology firm is planning its strategy for the next year and must choose exactly one of three possible options. One option is to innovate (Strategy A) by launching a new, cutting-edge product. Another option is to improve the current product line (Strategy B) by upgrading and refining what the firm already sells. The third option is to license an external technology (Strategy C) instead of developing new technology in-house.

The firm knows that future market demand is uncertain and can be in one of three states high demand, moderate demand or low demand. Based on market research, management estimates that the probability of high demand is 0.30, the probability of moderate demand is 0.40 and the probability of low demand is 0.30. These probabilities are treated as the best available information before any decision is made.

For each combination of strategy and demand, the firm has an estimate of the profit measured in thousands of dollars. If demand is high, the expected profits are 150 for Innovate (A), 110 for Improve (B) and 70 for License (C). If demand is moderate, the expected profits are 80 for Innovate (A), 90 for Improve (B) and 70 for License (C). If demand is low, the expected profits are -20 for Innovate (A), 40 for Improve (B) and 60 for License (C). The firm wants to analyse this decision using several decision-making rules, including expected value, maximax, maximin and the maximum opportunity minimax regret criterion, and then determine which strategy provides the most coherent choice given its view of risk and the available probability information.

1.2 Solution

1.2.1 (C2) Interpreting the parts of the decision problem

We identify the key elements and summarise them in a table.

Element	Description
Decision alternatives	Innovate (A), Improve (B), License (C)
States of nature	High demand, Moderate demand, Low demand
Fortuitous events	The actual market demand that will occur, which is unknown when deciding
Consequences	Profit in thousands of dollars for each strategy–demand combination
Probabilities	High demand 0.30, Moderate demand 0.40, Low demand 0.30

1.2.2 (C3) Building the payoff table from the description

From the problem description, we construct the payoff table (profits in thousands of dollars).

State of Nature	Probability	Innovate (A)	Improve (B)	License (C)
High demand	0.30	150	110	70
Moderate demand	0.40	80	90	70
Low demand	0.30	-20	40	60

1.2.3 (C4) Maximax rule decision making without probabilities

We ignore probabilities and focus only on the best possible payoff for each strategy.

Best payoff for each strategy

$$\max_i \text{Payoff}(A, S_i) = 150k$$

$$\max_i \text{Payoff}(B, S_i) = 110k$$

$$\max_i \text{Payoff}(C, S_i) = 70k$$

The largest of these is $150k$, so the Maximax rule recommends *Innovate (A)*.

1.2.4 (C5) Maximin rule decision making without probabilities

Now we ignore probabilities and focus on the worst-case payoff of each strategy.

Worst payoff for each strategy

$$\min_i \text{Payoff}(A, S_i) = -20k$$

$$\min_i \text{Payoff}(B, S_i) = 40k$$

$$\min_i \text{Payoff}(C, S_i) = 60k$$

We then pick the strategy with the highest of these worst outcomes

$$\max\{-20, 40, 60\} = 60k$$

The Maximin rule recommends *License (C)*.

1.2.5 (C6) Maximum opportunity criterion minimax regret

We construct the regret (opportunity loss) table. For each state of nature, we compare each payoff with the best payoff in that state.

Best payoff in each state

$$\text{High demand : } \max\{150, 110, 70\} = 150k$$

$$\text{Moderate demand : } \max\{80, 90, 70\} = 90k$$

$$\text{Low demand : } \max\{-20, 40, 60\} = 60k$$

Regrets are computed as best payoff minus actual payoff.

State of Nature	Innovate (A)	Improve (B)	License (C)
High demand	$150 - 150 = 0$	$150 - 110 = 40$	$150 - 70 = 80$
Moderate demand	$90 - 80 = 10$	$90 - 90 = 0$	$90 - 70 = 20$
Low demand	$60 - (-20) = 80$	$60 - 40 = 20$	$60 - 60 = 0$

Maximum regret for each strategy

$$\max_i \text{Regret}(A, S_i) = \max\{0, 10, 80\} = 80$$

$$\max_i \text{Regret}(B, S_i) = \max\{40, 0, 20\} = 40$$

$$\max_i \text{Regret}(C, S_i) = \max\{80, 20, 0\} = 80$$

The minimax regret rule chooses the strategy with the smallest maximum regret, so the maximum opportunity criterion recommends *Improve (B)*.

1.2.6 (C7) Using probabilities and expected value Bayes decision rule

We apply the Expected Monetary Value (EMV) rule. For each strategy, we multiply each payoff by the probability of its state and add the results.

For Innovate (A)

$$\begin{aligned} EMV_A &= 0.30(150) + 0.40(80) + 0.30(-20) \\ &= 0.30 \cdot 150 + 0.40 \cdot 80 + 0.30 \cdot (-20) \\ &= 45 + 32 - 6 = 71k \end{aligned}$$

For Improve (B)

$$\begin{aligned} EMV_B &= 0.30(110) + 0.40(90) + 0.30(40) \\ &= 0.30 \cdot 110 + 0.40 \cdot 90 + 0.30 \cdot 40 \\ &= 33 + 36 + 12 = 81k \end{aligned}$$

For License (C)

$$\begin{aligned} EMV_C &= 0.30(70) + 0.40(70) + 0.30(60) \\ &= 0.30 \cdot 70 + 0.40 \cdot 70 + 0.30 \cdot 60 \\ &= 21 + 28 + 18 = 67k \end{aligned}$$

Under the Bayes EMV rule, the best strategy is the one with the highest EMV, so EMV recommends *Improve (B)* with 81k.

1.2.7 (C9) Tree diagram of the decision-making problem

We represent the same situation as a decision tree.

Decision node choose strategy

Strategy A Innovate

High demand 0.30 → 150k
Moderate demand 0.40 → 80k
Low demand 0.30 → 20k

Strategy B Improve

High demand 0.30 → 110k
Moderate demand 0.40 → 90k
Low demand 0.30 → 40k

Strategy C License

High demand 0.30 → 70k
Moderate demand 0.40 → 70k
Low demand 0.30 → 60k

This diagram helps visualise all possible paths decisions followed by demand realisations and the associated payoffs.

1.2.8 (C8) Comparing Bayes, Maximin, Maximax and minimax regret

Summary of recommendations from the different rules

- Bayes EMV expected value with probabilities, no experimentation recommends *Improve (B)*.
- Maximum opportunity minimax regret recommends *Improve (B)*.
- Maximax very optimistic, ignores probabilities recommends *Innovate (A)*.
- Maximin very cautious, ignores probabilities recommends *License (C)*.

Bayes and minimax regret use the full payoff structure (and, in the case of Bayes, the probabilities) to favor a more balanced option, whereas Maximax and Maximin focus only on the most extreme possible outcomes.

1.2.9 (C10) Evaluating all possibilities in the decision problem

Considering all the strategies and rules together, the firm can reflect on the consequences of each possible choice. Innovate (A) offers the highest possible payoff but also the lowest worst-case payoff and a high maximum regret. License (C) protects strongly against the worst case but sacrifices expected profit and is not favoured by EMV or regret. Improve (B) is the only strategy that is simultaneously supported by expected value and by the minimax regret rule. This evaluation shows how different attitudes toward risk and regret lead to different recommended actions, and highlights Improve (B) as a compromise between risk and return.

1.2.10 (C1) Formulating the final decision and summarising all criteria

Based on each of the worked decision criteria, the recommended decision would be as follows.

If the firm decides according to the Maximax rule (C4) and is willing to act with a very optimistic, risk-seeking attitude, it should choose *Innovate (A)* because this strategy has the highest possible payoff among all options.

If the firm decides according to the Maximin rule (C5) and adopts a very conservative, worst-case oriented perspective, it should choose *License (C)* since this strategy has the strongest worst-case payoff.

If the firm decides according to the maximum opportunity minimax regret rule (C6) and wants to minimise the largest possible regret, it should choose *Improve (B)* because this strategy has the smallest maximum regret.

If the firm decides according to the Bayes EMV rule (C7) using the estimated probabilities and focusing on long-run average profit, it should again choose *Improve (B)* because it has the highest expected monetary value.

The decision tree (C9) does not change the recommendation but confirms, in a visual and systematic way, the same payoffs and probabilities used in these rules.

Finally, the comparison in (C8) and the global evaluation in (C10) show that each rule corresponds to a particular attitude toward risk and uncertainty.

Taking all criteria together, a coherent summary conclusion is that the strategy *Improve (B)* provides the most balanced decision for this firm. It is favoured by the expected value approach and by the minimax regret criterion, it avoids the very poor worst case of *Innovate (A)* and it still offers better long-run performance than *License (C)*. Therefore, in the context of this problem and given the available information, the firm should formulate its final decision as adopting the strategy *Improve (B)* for the next year.

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